

Hospital Management System: Normalization Report

1. First Normal Form (1NF) To achieve 1NF, we ensured that every table has a Primary Key to uniquely identify records, and that all columns contain atomic (indivisible) values. For example, in the **Patients** table, we split the patient's name into **FirstName** and **LastName** instead of a single "FullName" column.

2. Second Normal Form (2NF) To achieve 2NF, we ensured the database is in 1NF and that all non-key attributes are fully functionally dependent on the primary key. We avoided partial dependencies. For instance, the **Appointments** table only contains data specific to the appointment itself (Date, Status), while patient-specific details remain in the **Patients** table linked via the **PatientID** Foreign Key.

3. Third Normal Form (3NF) To achieve 3NF, we ensured the database is in 2NF and removed any transitive dependencies (where a non-key column depends on another non-key column). For example, in the **Billing** table, we only store the **TotalAmount** and **PaymentStatus**. We do not store the Doctor's specialization or the Patient's name in the billing table, as those can be retrieved via the **AppointmentID** foreign key relationship, preventing data anomalies and redundancy.